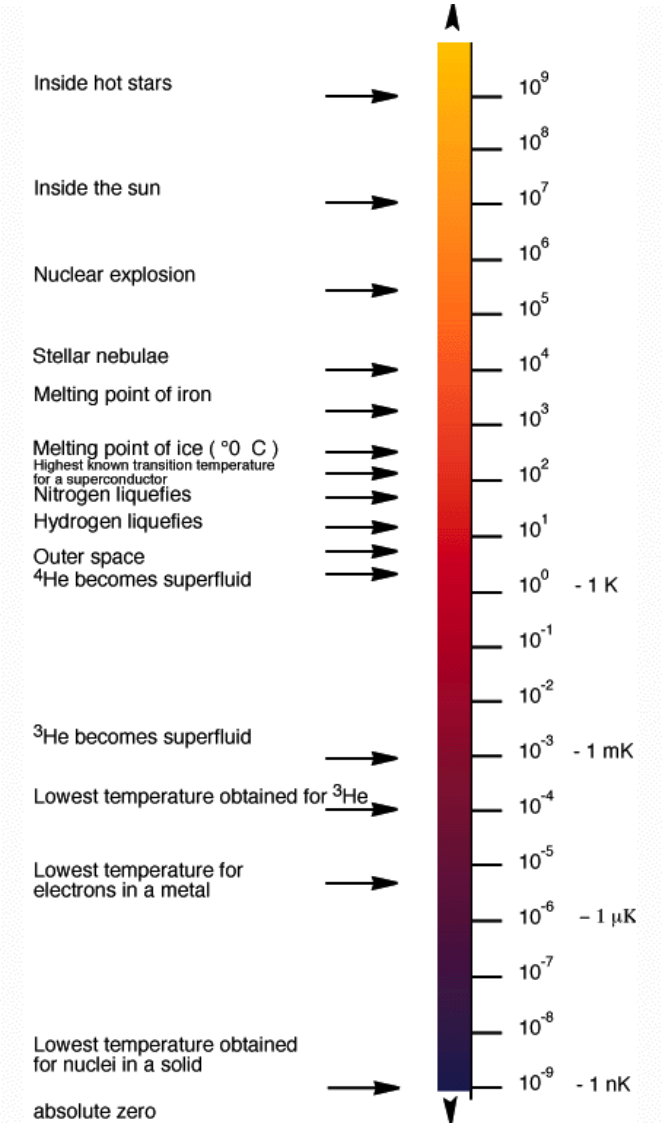
Temperature Measurement

Temperature Measurement

- Temperature measurement is a crucial part of many industrial processes
- Examples of industries where it is important are mineral processing, plastics, petrochemical and food
- There are a large number of different methods to measure temperature
- These use different physical properties. We will discuss some of these and look at some common temperature measurement sensors



wps.prenhall.com/esm_chaisson_astronomytoday (2007)

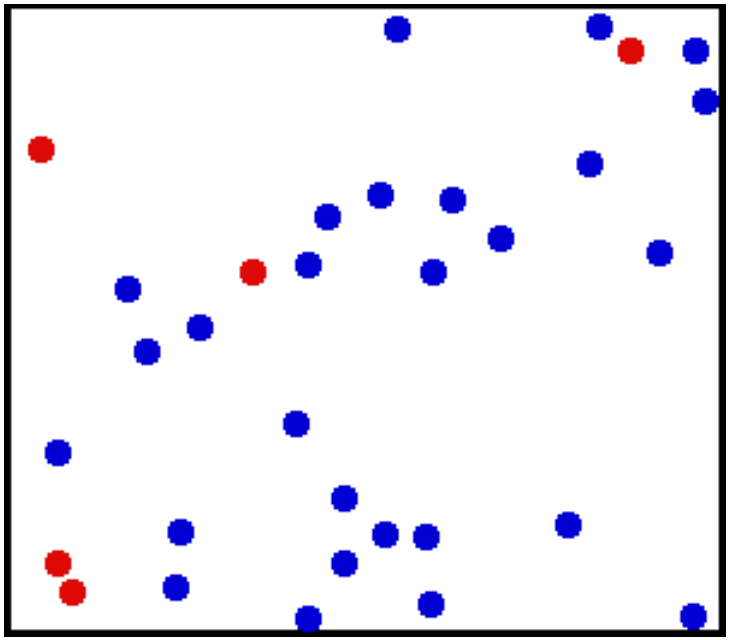
What is Temperature?

- Temperature is a measure of the average kinetic energy of particles in a medium

- The international unit for temperature is Kelvin (K) or degrees Celsius (ºC) where

$$K = °C + 273.15$$

- The measurement of low to medium temperatures (-273 ºC - ~500 ºC) is defined as thermometry while the measurement of higher temperatures it is known as pyrometry



Temperature Units?

- Fahrenheit. Early 1700s
- Celsius & Centrigrade. Mid 1700s
- Rankine. Mid 1800s
- Kelvin. Late 1800s

To convert °F to °C

$$°C = (°F - 32)5/9$$

To convert °C to °F

$$°F = (°C \times 9/5) + 32$$

To convert °F to °R

$$°R = °F + 459.6$$

To convert °C to K

$$K = °C + 273.15$$

To convert K to °R

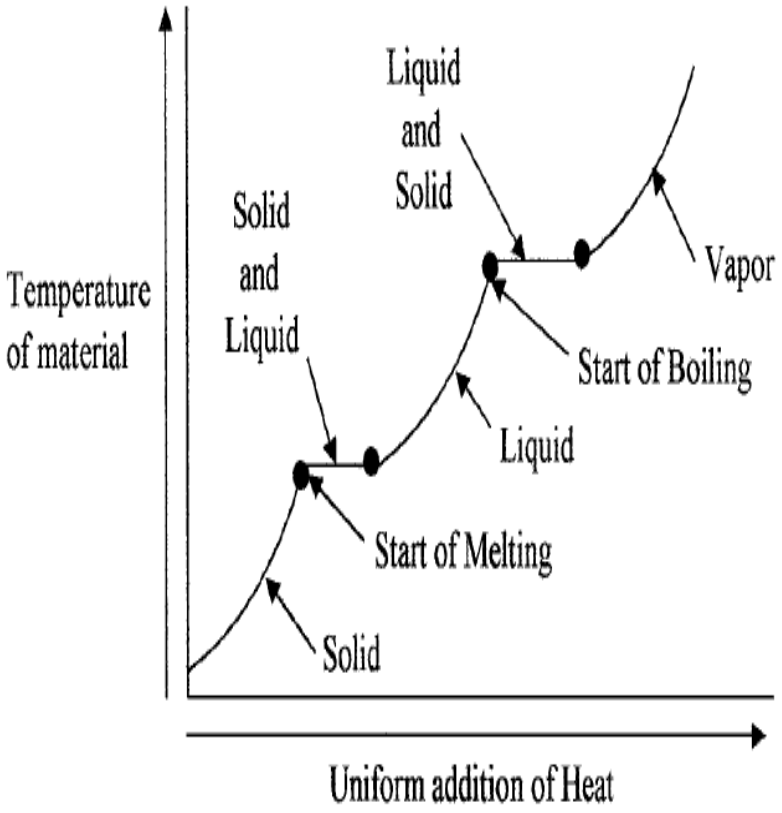
$$°R = 1.8 \times K$$

To convert °R to K

$$K = 0.555 \times °R$$

What is heat?

- Form of energy
- Energy cause vibration of molecules when applied to the system
- Temperature increases
- Heat Units
  - British Thermal Unit (BTU)
  - Calories
  - Joules
- Materials transition
- Specific heat
  - Quantity of heat energy required to raise the temp. of material by 1 degree celsius
- Methods of heat transfer



Thermal Measurement Method:

Linear Expansion of a Solid

- Rod thermometers and bimetallic thermometers are based on this principle

- These indicate temperature due to the different thermal expansion of two different metals

- A solid bar will change in length when it experiences a change in temperature

$$dL = \alpha L dT$$



www.moellerinstrument.com/moshock.shtml (2007)

- Where α is the linear temperature coefficient, *L* is the length of the bar and *dT* is the change in temperature

- If the original length of the bar is L_0 at T_0 , the new length L_1 at T_1 can be calculated as follows:

$$L_1 = L_0 + L_0\alpha(T_1 - T_0)$$
$$= L_0(1 + \alpha\Delta T)$$

Thermal Measurement Method:
Thermal Expansion of Liquids

- Used in liquid glass thermometers for a direct indication of temperature
- The principle is similar to that in solids except that we consider a volumetric temperature coefficient β

$$V_1 = V_0(1+\beta\Delta T)$$

- It is considered constant over a limited range



www.weissinstruments.com/catalogs/phc/sixhot.htm (2007)

Thermal Measurement Method:
Vapour Pressure of Liquids

- Vapour pressure is dependant on temperature
- The equation for an ideal gas is

$$pv = RT$$

- Where *p* is pressure, *v* is a specific volume, *T* is the temperature and *R* is the molar gas constant
- Therefore temperature can be measured in two ways:

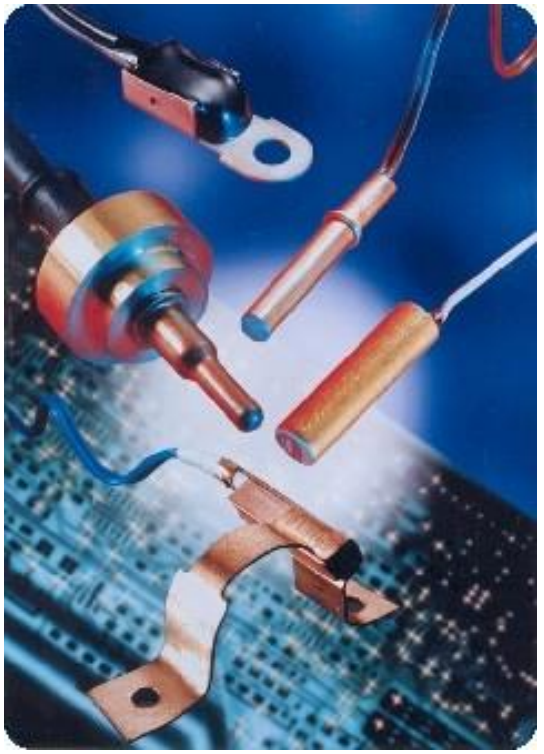


www.physics.umd.edu/.../demos/demosi3/i3-52.htm (2007)

  - Measuring the volume change at a constant pressure
  - Measuring the pressure difference at a constant volume

Temperature Measurement with Electrical Sensors

- Convert temperature to an electrical signal
- They often require some form of power source
- Great advantage is that the signals from these sensors are transmittable over long distances which makes remote measurement feasible



www.acal-radiatron.com/jamestech2.php (2007)

Conductivity in Metals

- Good conductivity in metals is due to the freely mobile electrons in the atomic lattice
- The number of free electrons and their kinetic energy are functions of temperature. As the temperature increases, the amplitude and frequency of vibration increases
- The free electrons' movement is now hindered through the medium and therefore the resistance of the material increases
- If an increase in temperature causes an increase in resistance of a material it is said to have a Positive Temperature Coefficient (PTC)

Conductivity in Metals

- The relationship between the temperature of metals and its electrical resistance is not liner but can be described by the following equation:

$$\Delta T = T_1 - T_0$$

$$R_1(t) = R_0\left(1 + a \cdot \Delta T + b \cdot \Delta T^2 + c \cdot T^3 + ...\right)$$

- Where R_1 is the resistance at temperature T_1, R_0 is the resistance of the material at a reference temperature T_0 .

- *a*, *b* and *c* are the temperature coefficients of resistivity and are dependant on the metal. They are only constant over a specific range

- However, for certain materials it is possible to neglect the higher terms for specific temperature ranges without introducing too large an error

- This reduces the equation to a linear relationship

Conductivity in Metals

- Any metal used for temperature measurement should meet these requirements:
  - Good long term stability in terms of resistance
  - High temperature coefficient of resistivity *a*
  - Resistant to corrosion and chemical impurities
  - Not effected by other physical quantities such as pressure
  - Good reproducibility of change in resistance as a function of temperature
- Platinum (Pt) and Nickel (Ni) satisfy most of the above requirements



www.matthey.com (2007)

Temperature Measurement:
Thermometers

- Mercury in glass
- Coefficient of expansion of mercury far greater than that of glass

- Advantages
  - Does not wet glass
  - Has a wide operating range
  - Clearly visible

- Disadvantages
  - Mercury is toxic
  - Easy of breakages



www.thermaldevices.com/.../RTDs (2007)

Electrical Temperature Measurement: Resistance Temperature Detectors (RTDs)

- Use the fact that certain materials resistance changes in a predictable way with a change in temperature
- The are mainly made from metallic conductors and mostly of platinum
- They are becoming the temperature sensor of choice in industry for temperature measurements below 600 ºC

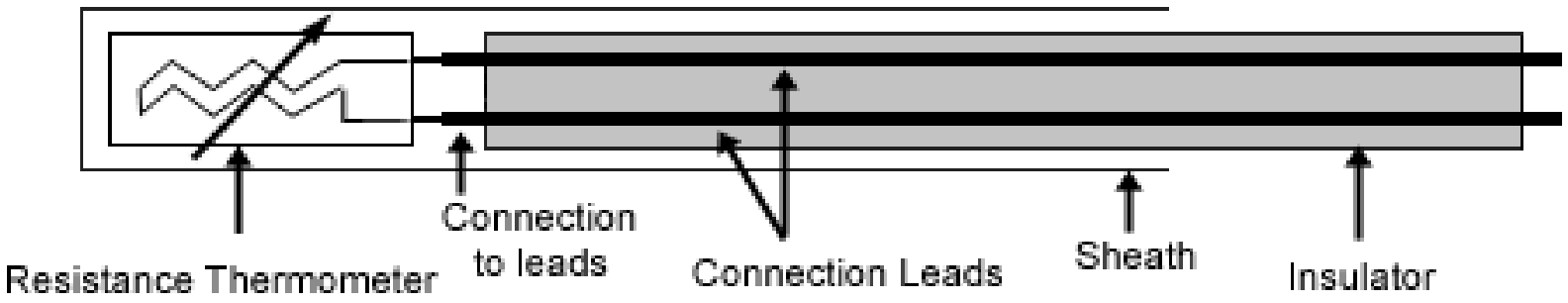


www.thermaldevices.com/.../RTDs (2007)

Resistance Temperature Detectors (RTDs)

- The most common types of RTD are:
  - Wire-wound in a ceramic insulator
  - Wires which are encapsulated in glass
- Resistance often measured using a Wheatstone Bridge arrangement

$$R_{T2} = R_{T1}(1+\alpha[T_2 - T_1])$$



www.wikipedia.com (2007)

Resistance Temperature Detectors (RTDs)

- Advantages:
  - High accuracy and can therefore be used in precision applications
  - Has low drift with time
  - Wide operating temperature range

- Disadvantages:
  - Are not often used above 660ºC as it is difficult to keep the platinum pure

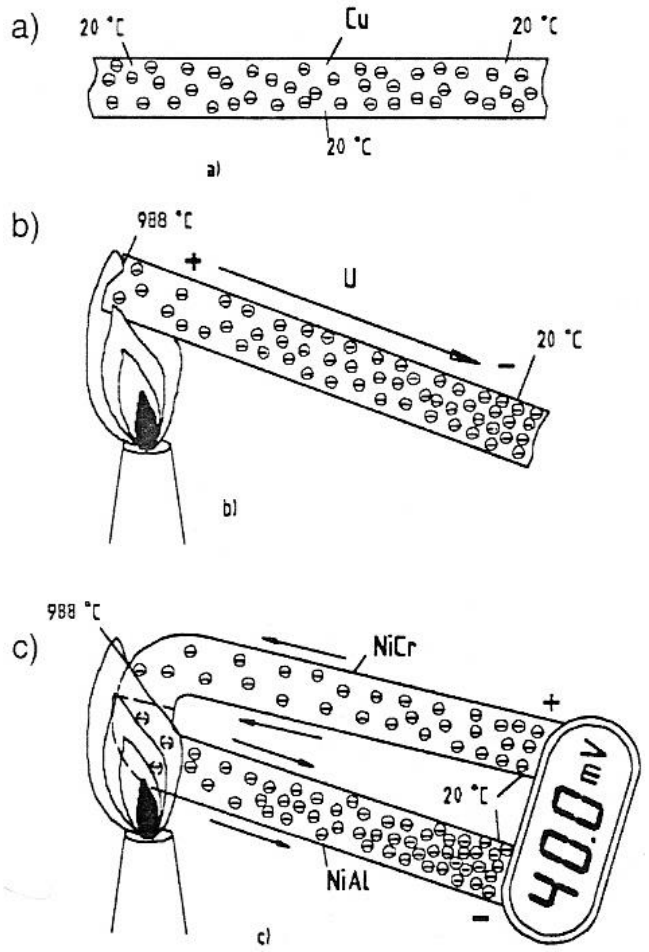
Thomas Johan Seebeck

- A German-Estonian Physicist who discovered that a voltage was produced across a metal bar when a temperature difference existed in the bar in 1821
- From this he formulated the Seebeck Principle which is used in some temperature measurement devices



The Seebeck Effect

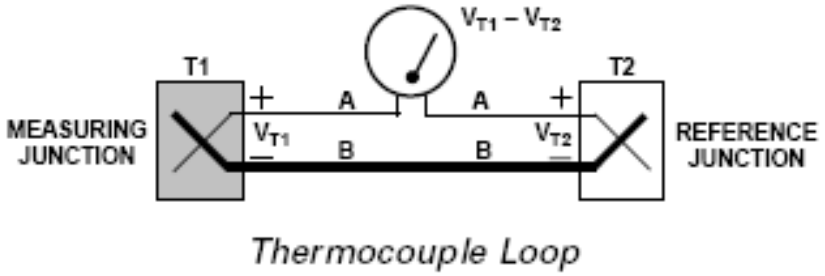
- If two different metals are joined together to form a continuous loop and their junctions are at different temperatures, an e.m.f. will be generated which cause a current to flow
- If a millivoltmeter is inserted into the loop, its output reading will give us an indication of the temperature difference between the two junctions of the loop
- This concept forms the basis of a thermocouple



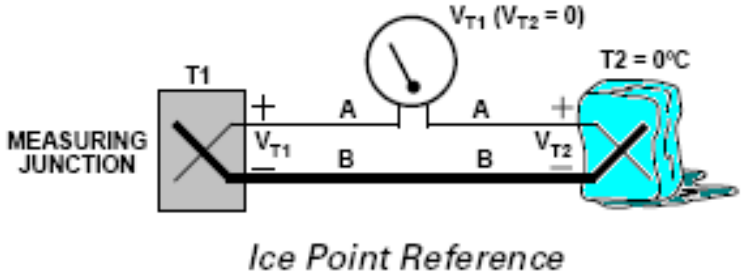
Lamour, R.. Lecture Notes (2003)

Electrical Temperature Measurement: Thermocouples

- Are the most commonly used electronic temperature measurement devices
- Consists of two dissimilar metals which are joined together at both ends
- One of the conductors is broken in the middle. A potential difference is generated across the break if the junctions are held at different temperatures
- Therefore if one end of a thermocouple is held at a known reference, the temperature of the other end can be calculated



Thermocouple Loop



Ice Point Reference

Analog Devices (1998) – AN369 Thermocouple Signal Conditioning using the AD594/AD595

Thermocouples Types

- Various metal combinations can be used for different temperature and voltage ranges, the following are examples of common combinations:

ANSI CODE	ALLOY COMBINATION	MAXMIUM TEMPERATURE RANGE	mV OUTPUT
B	*Platinum/Rhodium*	0°C to +1700°C	0 to +12.426
E	*Chromel/Constantan*	–200°C to +900°C	–8.824 to +68.783
J	*Iron/Constantan*	0°C to +750°C	0 to +42.283
K	*Chromel/Alumel*	–200°C to +1250°C	–5.973 to +50.633
N	*Nicrosil/Nisil*	–270°C to +1300°C	–4.345 to +47.502
R	*Platinum/Rhodium Platinum*	0°C to +1450°C	0 to +16.741
S	*Platinum/Rhodium Platinum*	0°C to +1450°C	0 to +14.973
T	*Copper/Constantan*	–200°C to +350°C	–5.602 to +17.816

Thermocouples

- Advantages:
  - Wide operating temperature range can be used at high temperatures
  - Fairly cheap
  - Interchangeable
  - Have standard connectors

- Disadvantages:
  - Lack of precision

Electrical Temperature Measurement: Semiconductor Sensors

- Silicon Measuring Resistors (PTC)
  - Small non-linearity
  - -70ºC - 160ºC gives a resistance change of 14Ω to 4kΩ

- Semiconductor Diodes
  - If supplied with a constant current, the conducting voltage is a function of absolute temperature
  - Almost linear between -50ºC - 150ºC

Electrical Temperature Measurement: Semiconductor Sensors

- Advantages
  - Lower cost
  - Accurate measuring device (due to its sensitivity)
- Disadvantages
  - Non-linearity over a wide temperature rannge

Electrical Temperature Measurement: Radiation Thermometers

- Also known as pyrometers
- They are non-contact sensors
- Used in the measurement range -100 ºC - 3500 ºC
- They are used to measure the temperature of a surface if it is visible
- Often used for objects with rapid temperature changes, moving objects and small objects



www.palloys.com.au/webcontent18.htm (2007)

Electrical Temperature Measurement: Capacitive and Inductive Sensors

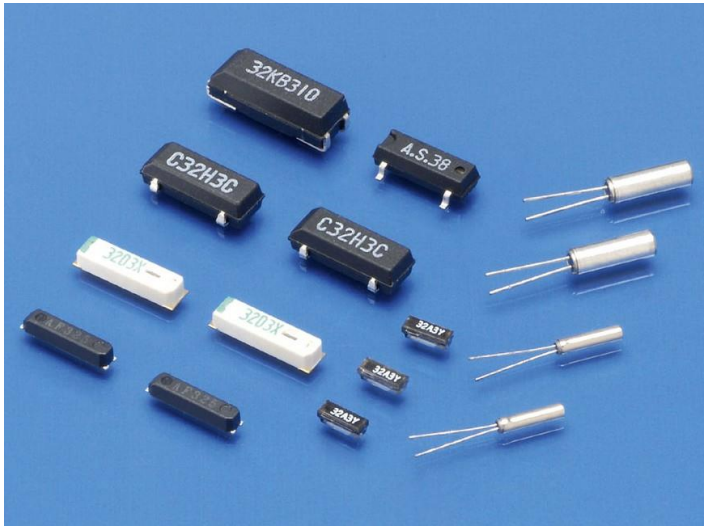
- Both these types of sensors are used in oscillating circuits
- The dielectric constant of most materials is temperature dependant
- The magnetic permeability of Vicalloy toroid coils changes as a function of temperature



www.tightpaktoroid.com/ (2007)

Electrical Temperature Measurement: Crystal Oscillator Methods

- If crystals are cut along certain axes, their resonant frequency will be largely affected by temperature
- The crystal can then be used in an oscillating circuit



www.sii.com.sg (2007)

Electrical Temperature Measurement: Temperature Dependant Colours

- Strips which change colour according to the temperature of the surface they are attached to
- For example the strips used on the side of fish tanks



www.omega.com (2007)